

Multinomial Logistic Regression. Softmax. Cross-Entropy

In case of two classes $y = \pm 1$, we introduced the following model:

$$p_w(x) = P(Y = 1 | X = x) = \sigma(w_0 + w_1x_1 + \dots + w_dx_d) = \frac{e^{w^T \tilde{x}}}{1 + e^{w^T \tilde{x}}}.$$

$$P(Y = -1 | X = x) = 1 - p_w(x) = \frac{1}{1 + e^{w^T \tilde{x}}}.$$

The linear function can be obtained as

$$w^T \tilde{x} = \log \frac{P(Y = 1 | X = x)}{P(Y = -1 | X = x)}.$$

It is called *logit* or *log-odds*. We could find the weights $w_0, w_1, \dots, w_d$ maximizing probability

$$L(w; Y^{(1)} = y^{(1)}, \dots, Y^{(N)} = y^{(N)} | X^{(1)} = x^{(1)}, \dots, X^{(N)} = x^{(N)}) = \prod_{i=1}^N (p_w(x^{(i)}))^{[y^{(i)}=1]} (1 - p_w(x^{(i)}))^{[y^{(i)}=-1]}$$

that is equivalent to minimizing log-loss

$$Loss(w) = \sum_{i=1}^N \log(1 + e^{-y^{(i)} w^T \tilde{x}}) \rightarrow \min_w.$$

If we have a lot of data, we can use a different approach. We could use two linear models, one for each class/probability:

$$a_1(x) = w_{1,0} + w_{1,1}x_1 + \dots + w_{1,d}x_d = w_1^T \tilde{x},$$

$$a_2(x) = w_{2,0} + w_{2,1}x_1 + \dots + w_{2,d}x_d = w_2^T \tilde{x}.$$

We can convert these numbers into probabilities as follows:

$$P(Y = 1 | X = x) = p_{w_1}(x) = \frac{e^{w_1^T \tilde{x}}}{e^{w_1^T \tilde{x}} + e^{w_2^T \tilde{x}}}, \quad P(Y = -1 | X = x) = p_{w_2}(x) = \frac{e^{w_2^T \tilde{x}}}{e^{w_1^T \tilde{x}} + e^{w_2^T \tilde{x}}}.$$

Problem 1. For a given x consider the following random variables:

$Y^{(i)} X^{(i)} = x^{(i)}$	-1	1
P	$p_{w_2}(x^{(i)})$	$p_{w_1}(x^{(i)})$

Write likelihood function $L(w_1, w_2)$ and log-loss used to find $w_{1,0}, w_{1,1}, \dots, w_{1,d}, w_{2,0}, w_{2,1}, \dots, w_{2,d}$.

